Importance of Cryptography: HTTP vs HTTPS Credential Transmission Main

1. OBJECTIVE

To investigate how dummy login credentials are transmitted over HTTP and HTTPS using Wireshark and compare the captured packets to explain how encryption in HTTPS protects sensitive information from being read by attacker.

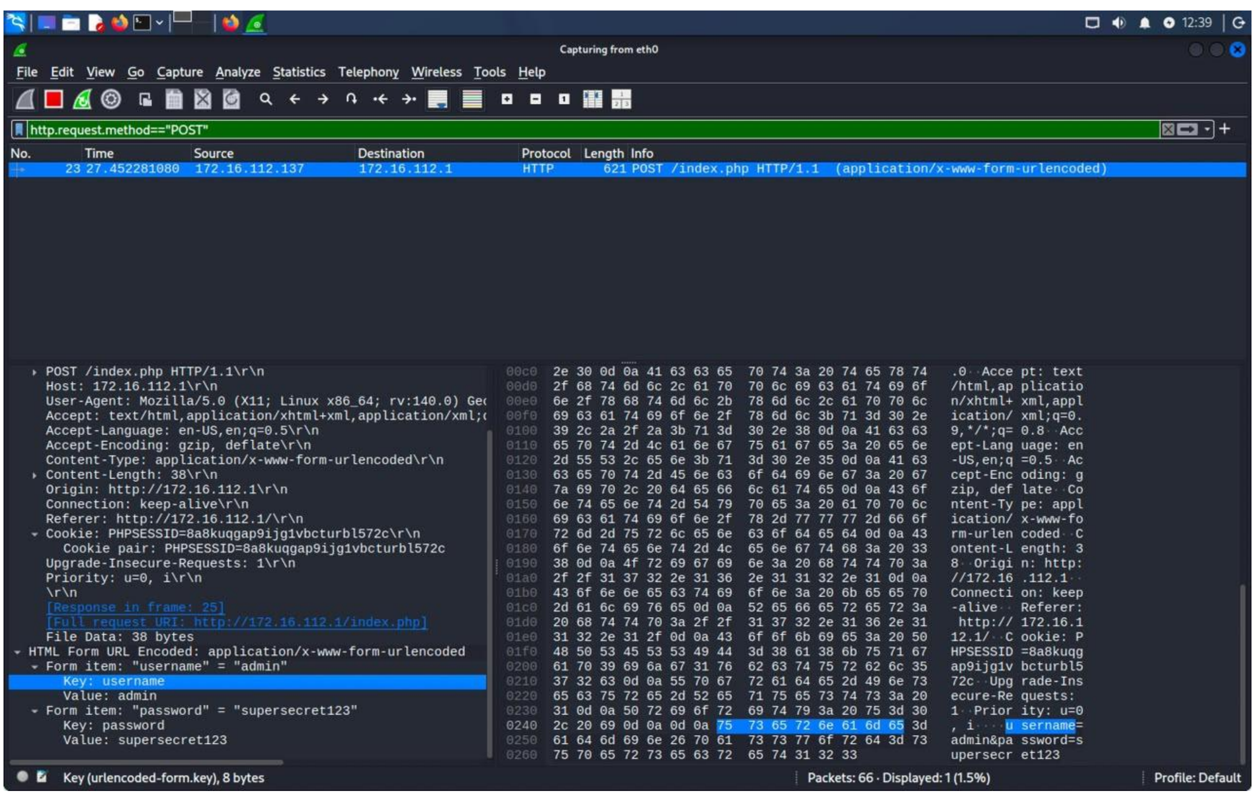
2. TOOLS USED

- Wireshark
- Web browser
- HTTP test login website
- GitHub login page

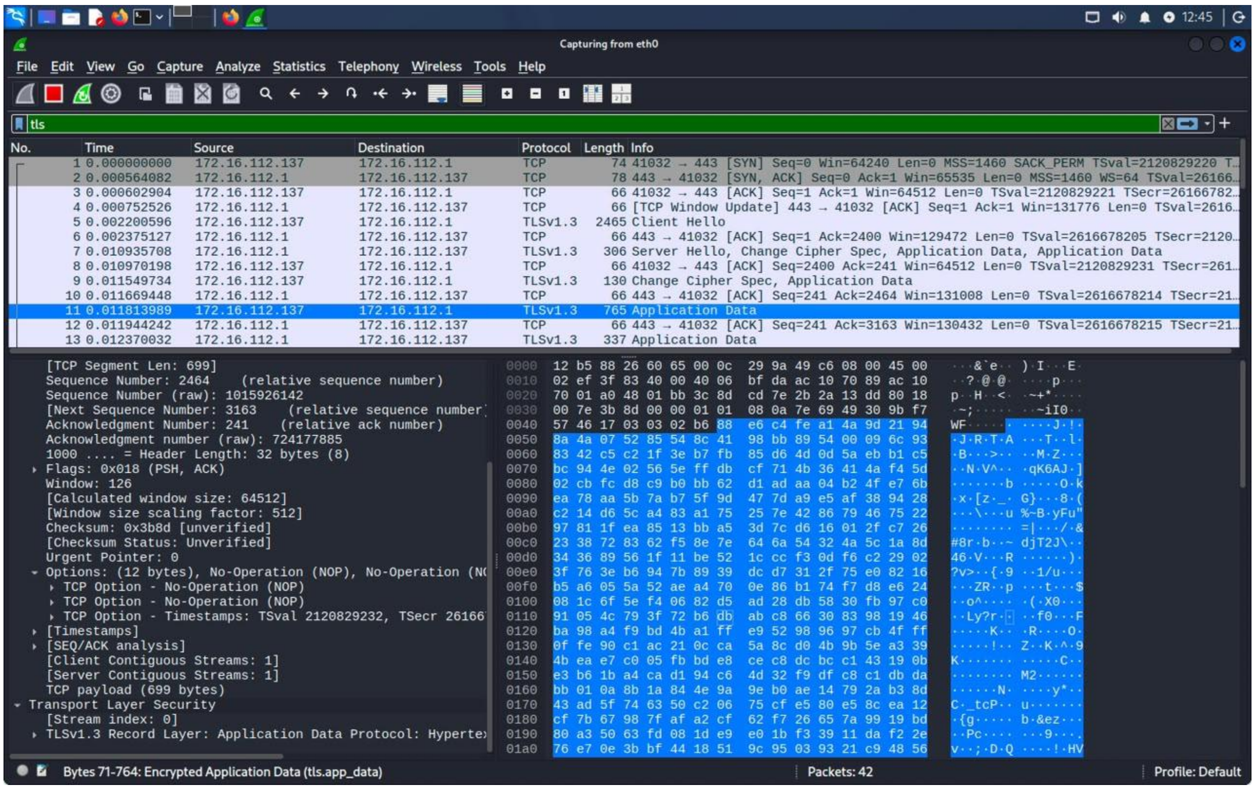
3. PROCEDURE

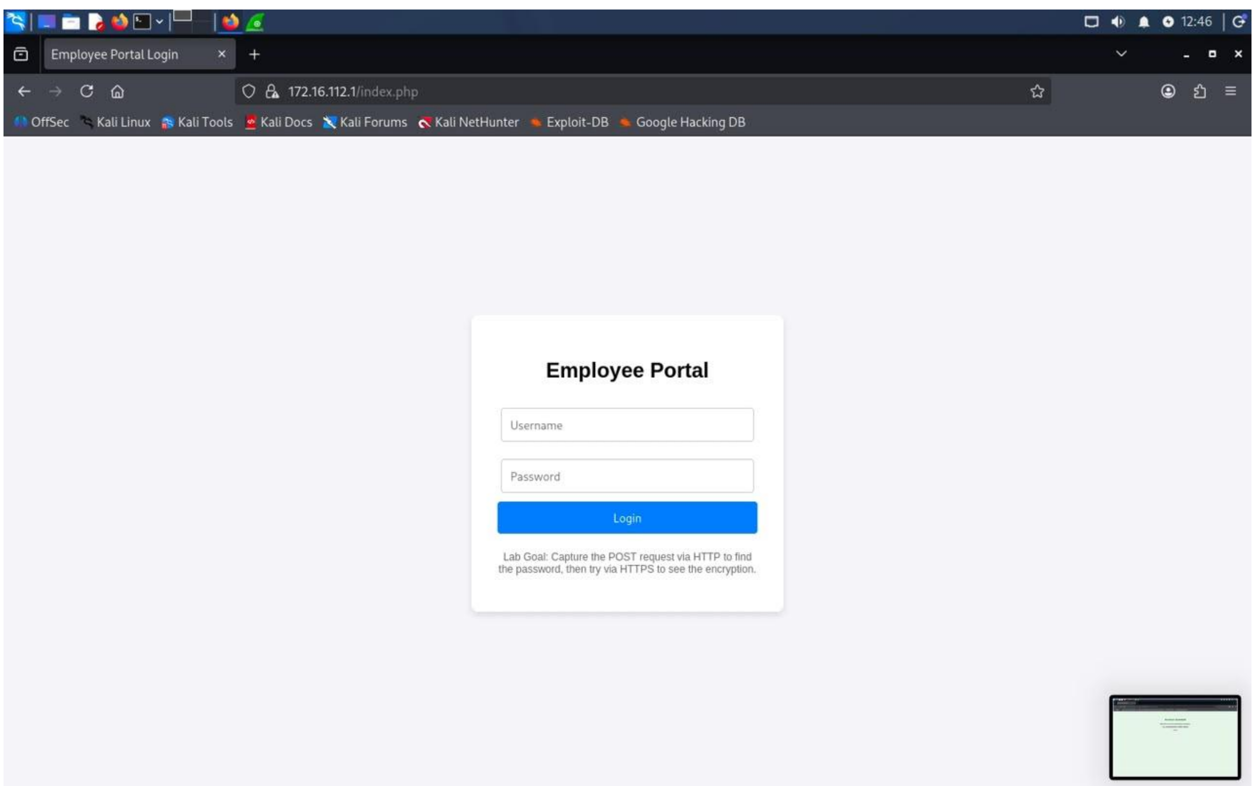
Wireshark was used to capture network traffic during two login experiments, one using an HTTP test login page and the other using an HTTPS secured.

HTTP Login: A packet capture was started on the active eth0 network interface before opening an HTTP test login website. Dummy login credentials were entered and submitted, after which the capture was stopped. The captured HTTP POST request was examined, and the username and password were visible in plain text under the HTML form data.



HTTPS Login: A new packet capture was started on the same network interface before opening the GitHub login page, which uses HTTPS. After entering dummy login credentials and submitting the form, the capture was stopped and the packets were analyzed. Unlike HTTP, the login credentials were not visible in readable form because the communication was protected by TLS encryption, which encrypts the data automatically during transmission.





4. COMPARISON

Features	HTTP	HTTPS
Data Transmission	Data is transmitted without encryption, so login information may be visible in Wireshark.	Data is encrypted using TLS, so the actual login information cannot be directly read.
Security & Confidentiality	Does not provide confidentiality, making sensitive information vulnerable to interception.	Provides confidentiality and better protection for sensitive information during transmission.
Use for Login	Not suitable for securely transmitting usernames and passwords.	Recommended for login pages and other communications involving sensitive information.

5. IMPORTANCE OF CRYPTOGRAPHY

Cryptography helps keep sensitive information safe when it is sent over a network. In this experiment, the login details sent through HTTP could be seen directly in Wireshark because they were not encrypted. When HTTPS was used, the login information was encrypted using TLS, so it could not be read directly from the captured packets. This shows how cryptography protects information from being easily accessed by someone monitoring the network. It also helps maintain the confidentiality and integrity of the data during transmission.

6. CONCLUSION

This experiment helped us understand the practical difference between HTTP and HTTPS. In the HTTP capture the dummy login details were visible in Wireshark while in the HTTPS capture the details were encrypted and could not be read directly. From this we can see how encryption helps protect sensitive information such as usernames and passwords when they are sent over a network.